

BIG DATA ANALYTICS LAB

IV B. TECH- I SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4CS20	PCC	L	T	P	C	CIE	SEE	Total
		-	-	4	2	30	70	100

COURSE OBJECTIVES:

To learn

1. To introduce the terminology, technology and its applications
2. To introduce the concept of Analytics for Business
3. To introduce the tools, technologies & programming languages this is used in day to day analytics cycle.

COURSE OUTCOMES:

Upon successful completion of the course, the student is able to

1. Connect to Hadoop cluster, experiment with various Linux and HDFS commands to store data.
2. Apply the knowledge of Map Reduce programming to process the stored data in HDFS.
3. Make use of database operations to store results in tables and generate reports.
4. Connect to web data sources for data gathering, Integrate data sources with Hadoop components to process streaming data.
5. Generate reports using data visualization tools.

SYLLABUS

Week 1

- i) Perform setting up and installing Vmware for Hadoop and Linux.
- ii) Basic Linux Commands

Week 2

Run basic HDFS shell commands

Week 3

Implement the following file management tasks in Hadoop:

- Adding files and directories
- Retrieving files
- Deleting files and directories.

Hint: A typical Hadoop workflow creates data files (such as log files) elsewhere and copies them into HDFS using one of the above command line utilities.

Week 4

Write the steps to export JAR using eclipse.

Run a basic Word Count Map Reduce program to understand Map Reduce Paradigm

Week 5

Write a Map Reduce program that mines weather data.

(Weather sensors collecting data every hour at many locations across the globe gather a large volume of log data, which is a good candidate for analysis with MapReduce, since it is semi structured and record-oriented).

Week 6

Run Pig and perform basic PIG commands.

Week 7

Write Pig Latin scripts to sort, group, join, project, and filter your data.

Week 8

Run HIVE and perform basic HIVE commands to create a table and enter data into tables.

Week 9

Use Hive to create, alter, and drop databases, tables, views, functions, and indexes

Week 10

Use CDH and HUE to analyze data and generate reports for sample datasets

Week 11

Importing and exporting Data in HDFS using Sqoop from MySQL database

Week 12

Use data visualization tool to generate reports on sample datasets.